Exercise 4 – Classification II

Introduction to Machine Learning

Hint: Useful libraries

R

```r
# you may need the following packages for this exercise sheet:

library(mlr3)
library(mlr3learners)
library(ggplot2)
library(mlbench)
library(mlr3viz)
```

Python

```python
# Consider the following libraries for this exercise sheet:

# general
import numpy as np
import pandas as pd
from scipy.stats import norm

# plotting
import matplotlib.pyplot as plt
import seaborn as sns

# sklearn
from sklearn.naive_bayes import (
    CategoricalNB,
)  # import Naive Bayes Classifier for categorial distributed features
from sklearn.naive_bayes import (
    GaussianNB,
)  # import Naive Bayes Classifier for normal distributed features
```

```python
from sklearn.preprocessing import OrdinalEncoder
from sklearn.preprocessing import LabelEncoder
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
from sklearn.discriminant_analysis import QuadraticDiscriminantAnalysis as QDA
from sklearn.inspection import DecisionBoundaryDisplay
from sklearn.metrics import confusion_matrix
from sklearn.metrics import precision_recall_fscore_support
```

Exercise 1: Naive Bayes

Learning goals

Compute Naive Bayes predictions by hand

You are given the following table with the target variable `Banana`:

ID	Color	Form	Origin	Banana
1	yellow	oblong	imported	yes
2	yellow	round	domestic	no
3	yellow	oblong	imported	no
4	brown	oblong	imported	yes
5	brown	round	domestic	no
6	green	round	imported	yes
7	green	oblong	domestic	no
8	red	round	imported	no

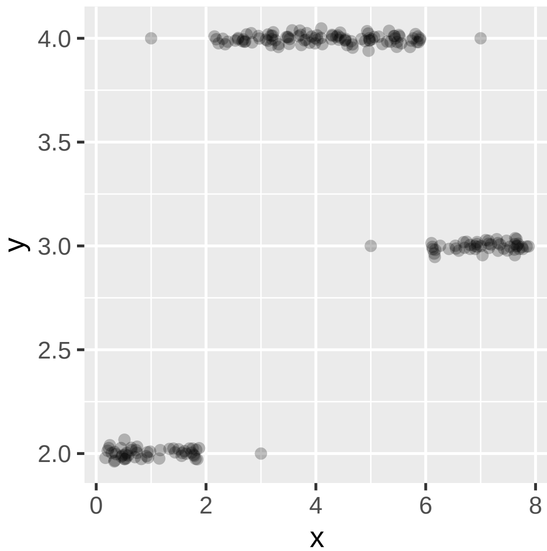
a) We want to use a Naive Bayes classifier to predict whether a new fruit is a `Banana` or not. Estimate the posterior probability $\hat{\pi}(\mathbf{x}_*)$ for a new observation $\mathbf{x}_* = (\text{yellow}, \text{round}, \text{imported})$. How would you classify the object?

b) Assume you have an additional feature `Length` that measures the length in cm. Describe in 1-2 sentences how you would handle this numeric feature with Naive Bayes.

Exercise 2: Discriminant analysis

Learning goals

1) Set up discriminant analysis by hand
2) Make predictions with discriminant analysis
3) Discuss difference between LDA and QDA



The above plot shows $\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \dots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right)$, a data set with $n = 200$ observations of a continuous target variable y and a continuous, 1-dimensional feature variable $\mathbf{x}$. In the following, we aim at predicting y with a machine learning model that takes $\mathbf{x}$ as input.

a) To prepare the data for classification, we categorize the target variable y in 3 classes and call the transformed target variable z, as follows:

$$z^{(i)} = \begin{cases} 1, & y^{(i)} \in (-\infty, 2.5] \\ 2, & y^{(i)} \in (2.5, 3.5] \\ 3, & y^{(i)} \in (3.5, \infty) \end{cases}$$

Now we can apply quadratic discriminant analysis (QDA):

i. Estimate the class means $\mu_k = \mathbb{E}(\mathbf{x}|z = k)$ for each of the three classes $k \in \{1, 2, 3\}$ visually from the plot. Do not overcomplicate this, a rough estimate is sufficient here.

ii. Make a plot that visualizes the different estimated densities per class.

b)

i. How would your plot from ii) change if we used linear discriminant analysis (LDA) instead of QDA? Explain your answer.

ii. Why is QDA preferable over LDA for this data?

c) Given are two new observations $\mathbf{x}_{*1} = -10$ and $\mathbf{x}_{*2} = 7$. Assuming roughly equal class sizes, state the prediction for QDA and explain how you arrive there.

Exercise 3: Decision boundaries for classification learners

The whole exercise is only for lecture group B!

Learning goals

Get a feeling for decision boundaries produced by LDA/QDA/NB

We will now visualize how well different learners classify the three-class `mlbench::mlbench.cassini` data set.

a)
- Generate 1000 points from `cassini` using R or import `cassini_data.csv` in Python.
- Then, perturb the `x.2` dimension with Gaussian noise (mean 0, standard deviation 0.5), and consider the classifiers already introduced in the lecture:
  - LDA (Linear Discriminant Analysis),
  - QDA (Quadratic Discriminant Analysis), and
  - Naive Bayes.

Plot the learners' decision boundaries. Can you spot differences in separation ability?

(Note that logistic regression cannot handle more than two classes and is therefore not listed here.)

Exercise 4: LDA and QDA as linear/quadratic classifiers

The whole exercise is only for lecture group A!

Learning goals

1. Derive why LDA has a linear decision boundary
2. Derive why QDA has a quadratic decision boundary
3. Connect the covariance assumption to the shape of the boundary

In discriminant analysis we model the class-conditional densities as multivariate Gaussians,

$$p(\mathbf{x}|y=k) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} \exp\left(-\tfrac{1}{2}(\mathbf{x}-\mu_k)^\top \Sigma_k^{-1}(\mathbf{x}-\mu_k)\right),$$

and obtain the posterior via Bayes' theorem, $\pi_k(\mathbf{x}) \propto \pi_k\, p(\mathbf{x}|y=k)$ (the denominator $p(\mathbf{x})$ is the same for all classes). Since we assign $\mathbf{x}$ to the class with the largest posterior and log is strictly increasing, it suffices to compare the **discriminant functions**

$$\delta_k(\mathbf{x}) := \log\left(\pi_k\, p(\mathbf{x}|y=k)\right).$$

The decision boundary between two classes j and k is the set of points where they are tied, $\{\mathbf{x} : \delta_j(\mathbf{x}) = \delta_k(\mathbf{x})\}$. Throughout, we may drop any term that does **not** depend on the class k: such terms are identical for all classes and cancel in the difference $\delta_j(\mathbf{x}) - \delta_k(\mathbf{x})$.

a) **LDA** assumes a shared covariance matrix $\Sigma_k = \Sigma$ for all classes. Show that $\delta_k(\mathbf{x})$ is affine-linear in $\mathbf{x}$, i.e. $\delta_k(\mathbf{x}) = w_{0k} + \mathbf{x}^\top w_k$ with a scalar intercept $w_{0k} \in \mathbb{R}$ and a coefficient vector $w_k = (w_{1k}, \dots, w_{pk})^\top \in \mathbb{R}^p$, and give w_{0k} and w_k. Conclude that the decision boundary between any two classes is a hyperplane.

   Hint

   Expand the quadratic form $(\mathbf{x}-\mu_k)^\top \Sigma^{-1}(\mathbf{x}-\mu_k)$ and check which of the resulting terms depend on the class k.

b) **QDA** drops this assumption: each class keeps its own covariance matrix Σ_k. Show that $\delta_k(\mathbf{x})$ now contains a term $-\frac{1}{2}\mathbf{x}^\top \Sigma_k^{-1}\mathbf{x}$ that depends on the class, so that δ_k is quadratic in $\mathbf{x}$ and the decision boundary is a curved (quadratic) surface.

c) Finally, we examine the cost of QDA's flexibility and how Naive Bayes relates to LDA and QDA.

   i. Recall which modeling assumption distinguishes QDA from LDA. Based on this, what does QDA have to estimate from the data that LDA does not, and how many additional free parameters does this require? What practical disadvantage for QDA follows?

   ii. In Exercise 1, Naive Bayes assumed the features to be conditionally independent given the class. For **numeric** features modeled by Gaussians, this makes Gaussian Naive Bayes a special case of QDA. What does the independence assumption imply for the covariance matrices Σ_k, and how does the shape of the class densities (and hence the decision boundary) then compare to a *general* QDA?

   Hint

   Recall what the off-diagonal entries of a covariance matrix encode.